Recording Activity: Multimeter and Ohm's Law

Name: ______________________________ Date: __________________

Student: Write predictions, teacher-approved readings, calculations, and explanations here. **Teacher:** Approve each measurement and complete every safety checkpoint in the lesson instructions. This worksheet does not give permission to change the meter mode or connect power without that approval.

Predict

Before measuring current, what do you predict the meter will show when S2 is released? What do you predict it will show when S2 is pressed?

Record in Lesson Order

Record only teacher-approved measurements. Fill the table from left to right as you complete the lesson. Do not prepare the meter for current mode until you have recorded the predicted current.

Switch or circuit state	R1 fixed resistance (Ω)	B1 battery voltage (V)	Predicted current (mA)	Measured current (mA)	Approximate comparison
S2 pressed; meter in series completes the path					Close / not close: __________

Show the calculation used for the predicted current:
__________ V ÷ __________ Ω = __________ A
__________ A × 1,000 = __________ mA predicted

Observe the Switch

Complete these readings only during the teacher-verified current procedure:

- With the meter in series and S2 released, the circuit is open. Meter reading: __________ mA.
- With the meter in series and S2 pressed, the circuit is closed. Meter reading: __________ mA.

Compare and Explain

1. What changed when the circuit went from open to closed?

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2. Were the measured and predicted current values approximately close? Describe the comparison; they do not need to be exactly equal.

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3. What evidence showed that current was flowing through the closed circuit?
